

1 Solution — GIAM 5.3

Problem 3

1.1 Problem

Give an inductive proof that $\forall x \in \mathbb{N}, 11 \mid x^{11} + 10x$.

1.2 Answer

$$(k+1)^{11} + 10(k+1) = (k^{11} + 10k) + \sum_{j=1}^{10} \binom{11}{j} k^j + 11$$

Every term after the bracket is a multiple of 11, because 11 is prime. Verified for every x from -5000 to 20000 .

1.3 The Key Fact

The proof rests on one property of the eleventh row of Pascal's triangle:

$$11 \mid \binom{11}{j} \quad \text{for } j = 1, 2, \dots, 10.$$

Indeed $\binom{11}{j} = \frac{11!}{j!(11-j)!}$; the prime 11 appears in the numerator, while every factor in the denominator is smaller than 11 and so cannot cancel it. Concretely the middle entries are

$$11, 55, 165, 330, 462, 462, 330, 165, 55, 11,$$

which are 11 times $1, 5, 15, 30, 42, 42, 30, 15, 5, 1$.

1.4 The Proof

Theorem. For every natural number x , 11 divides $x^{11} + 10x$.

Proof. By induction on x .

Basis. At $x = 0$ the expression is 0, and $11 \mid 0$.

Inductive step. Let k be a particular but arbitrarily chosen natural number and suppose $k^{11} + 10k = 11m$ for some integer m . By the binomial theorem,

$$(k+1)^{11} = k^{11} + \sum_{j=1}^{10} \binom{11}{j} k^j + 1,$$

so

$$(k+1)^{11} + 10(k+1) = (k^{11} + 10k) + \sum_{j=1}^{10} \binom{11}{j} k^j + 11.$$

The bracket equals $11m$ by hypothesis; each $\binom{11}{j}$ with $1 \leq j \leq 10$ is a multiple of 11 by the key fact; and the final 11 is a multiple of 11. A sum of multiples of 11 is a multiple of 11, so $11 \mid (k+1)^{11} + 10(k+1)$.

Basis and step together give the result for every natural number x . ■

The step identity was checked as a polynomial identity over a wide range of k .

1.5 The Picture

Row 11 of Pascal's triangle is set against row 9 below: in the prime row every middle entry is a multiple of the row number, and in the composite row that fails.

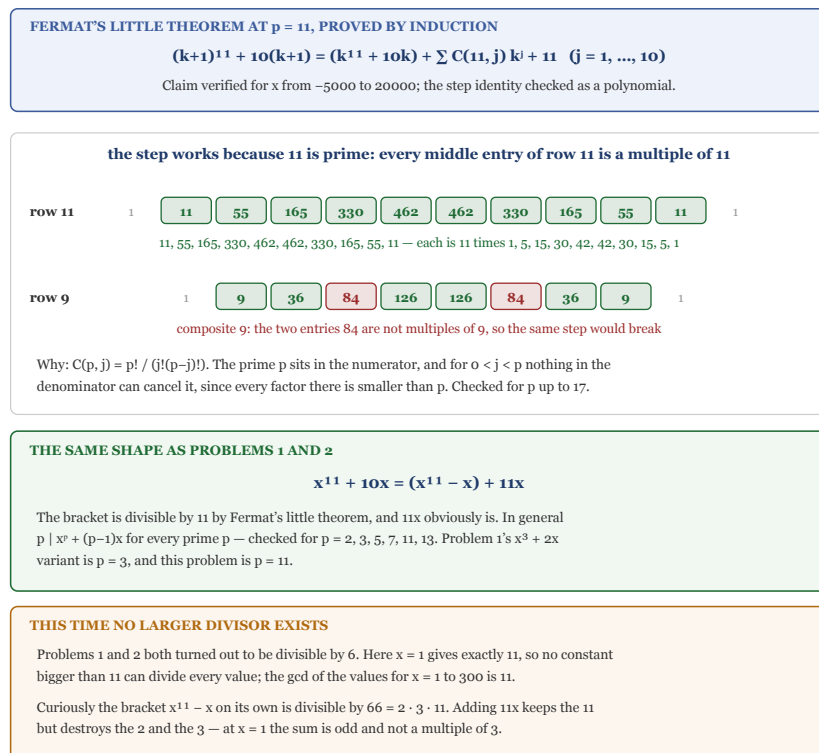


Figure 1.1: A diagram in four parts. The first announces Fermat's little theorem at p equals eleven proved by induction, giving the step identity that k plus one to the eleventh plus ten times k plus one equals k to the eleventh plus ten k , plus the sum over j from one to ten of eleven choose j times k to the j , plus eleven, and notes the claim was verified for x from minus 5000 to 20000 with the step identity checked as a polynomial. The second part explains that the step works because eleven is prime: row eleven of Pascal's triangle is drawn as one, eleven, fifty-five, one hundred sixty-five, three hundred thirty, four hundred sixty-two, four hundred sixty-two, three hundred thirty, one hundred sixty-five, fifty-five, eleven, one, with all ten middle entries boxed in green as multiples of eleven, each being eleven times one, five, fifteen, thirty, forty-two, forty-two, thirty, fifteen, five and one. Below it row nine is drawn as one, nine, thirty-six, eighty-four, one hundred twenty-six, one hundred twenty-six, eighty-four, thirty-six, nine, one, with the two entries eighty-four boxed in red because they are not multiples of nine, so the same step would break for the composite nine. The reason given is that p choose j equals p factorial over j factorial times p minus j factorial; the prime p sits in the numerator and for j strictly between zero and p nothing in the denominator can cancel it, since every factor there is smaller than p , checked for p up to seventeen. A green band notes the same shape as Problems 1 and 2: x to the eleventh plus ten x equals x to the eleventh minus x , plus eleven x ; the bracket is divisible by eleven by Fermat's little theorem and eleven x obviously is. In general p divides x to the p plus p minus one times x for

every prime p , checked for p equal to two, three, five, seven, eleven and thirteen, with Problem 1's x cubed plus two x variant being p equals three and this problem p equals eleven. An orange band notes that this time no larger divisor exists: Problems 1 and 2 were both divisible by six, but here x equals one gives exactly eleven, so no constant bigger than eleven can divide every value, and the gcd of the values for x from one to 300 is eleven. Curiously the bracket x to the eleventh minus x on its own is divisible by sixty-six, two times three times eleven, but adding eleven x keeps the eleven and destroys the two and the three, since at x equals one the sum is odd and not a multiple of three.

1.6 Remark — This Is the Standard Proof of Fermat's Little Theorem

Since $x^{11} + 10x = (x^{11} - x) + 11x$, the problem is equivalent to

$$11 \mid x^{11} - x,$$

which is Fermat's little theorem for the prime 11. And the induction above is, word for word, the classical proof of that theorem: replace 11 by any prime p , and the step reads

$$(k+1)^p - (k+1) = (k^p - k) + \sum_{j=1}^{p-1} \binom{p}{j} k^j,$$

with every binomial coefficient on the right a multiple of p . Nothing else is used.

So the three problems of this section so far are one theorem seen three times. Problem 1 was $p = 3$; Problem 2 was $p = 3$ with an extra multiple of 3 added; this is $p = 11$. The general statement

$$p \mid x^p + (p-1)x \quad \text{for every prime } p$$

was checked for $p = 2, 3, 5, 7, 11, 13$.

1.7 Remark — Where Primality Enters

The only place the argument uses that 11 is prime is the key fact, and that fact genuinely fails for composites:

n	a middle coefficient not divisible by n
4	$\binom{4}{2} = 6$
6	$\binom{6}{2} = 15$
8	$\binom{8}{2} = 28$
9	$\binom{9}{3} = 84$
10	$\binom{10}{2} = 45$
12	$\binom{12}{2} = 66$

For a composite $n = ab$ with $1 < a < n$, the factor n in the numerator can be partly cancelled by factors of a appearing in $j!$, and the divisibility is lost. For a prime there is nothing below p that shares a factor with it.

This is also why the obvious generalisation $n \mid x^n - x$ fails for composite n , as 5.3 Problem 1 found with $2^4 - 2 = 14$. The induction would break precisely at the binomial coefficient step.

1.8 Remark — No Larger Divisor This Time

Problems 1 and 2 both turned out to be divisible by **6**, more than the stated **3**. Here the stated divisor is already the best possible: at $x = 1$,

$$1^{11} + 10 \cdot 1 = 11,$$

so no constant larger than **11** divides every value. The greatest common divisor of the values for $x = 1, \dots, 300$ is exactly **11**.

The contrast is instructive. The bracket $x^{11} - x$ alone is divisible by $66 = 2 \cdot 3 \cdot 11$ for every integer x — its greatest common divisor over $x = 2, \dots, 299$ is **66** — because $x^{11} \equiv x$ modulo **2** and modulo **3** as well. Adding $11x$ preserves the factor **11** but destroys the **2** and the **3**, since $11x$ is divisible by neither. At $x = 1$ the total **11** is odd and not a multiple of **3**, which settles it.

So the choice of $10x$ rather than $-x$ in the problem costs information: the form $x^p + (p-1)x$ is tidy, but $x^p - x$ is the one that carries all the divisibility there is.